

Chapter 10: Optical Properties

- Optical properties of ceramic products are of concern in different applications:
 - Optical glass and crystals:** windows, lenses, prisms, filters...;
 - Color, translucency, and surface gloss:** tiles, ceramic tableware, artware, porcelain enamels, sanitary wares.....

ISSUES TO ADDRESS...

- What phenomena occur when light is shined on a ceramic?
- What determines the characteristic colors of ceramics?
- Why are some materials transparent and others are translucent or opaque?
- How does a laser operate?

The nature of light

Light has both particulate and wavelike characteristics

📄 **Photon** - a quantum unit of light

$$E = h\nu = \frac{hc}{\lambda}$$

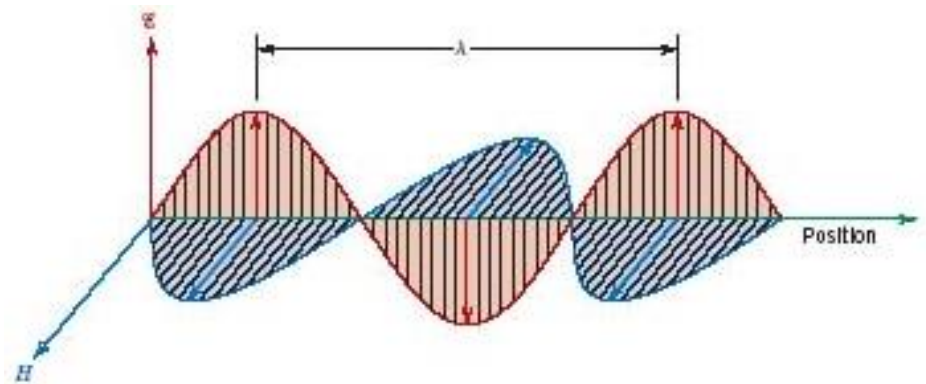
E = energy of a photon

λ = wavelength of radiation

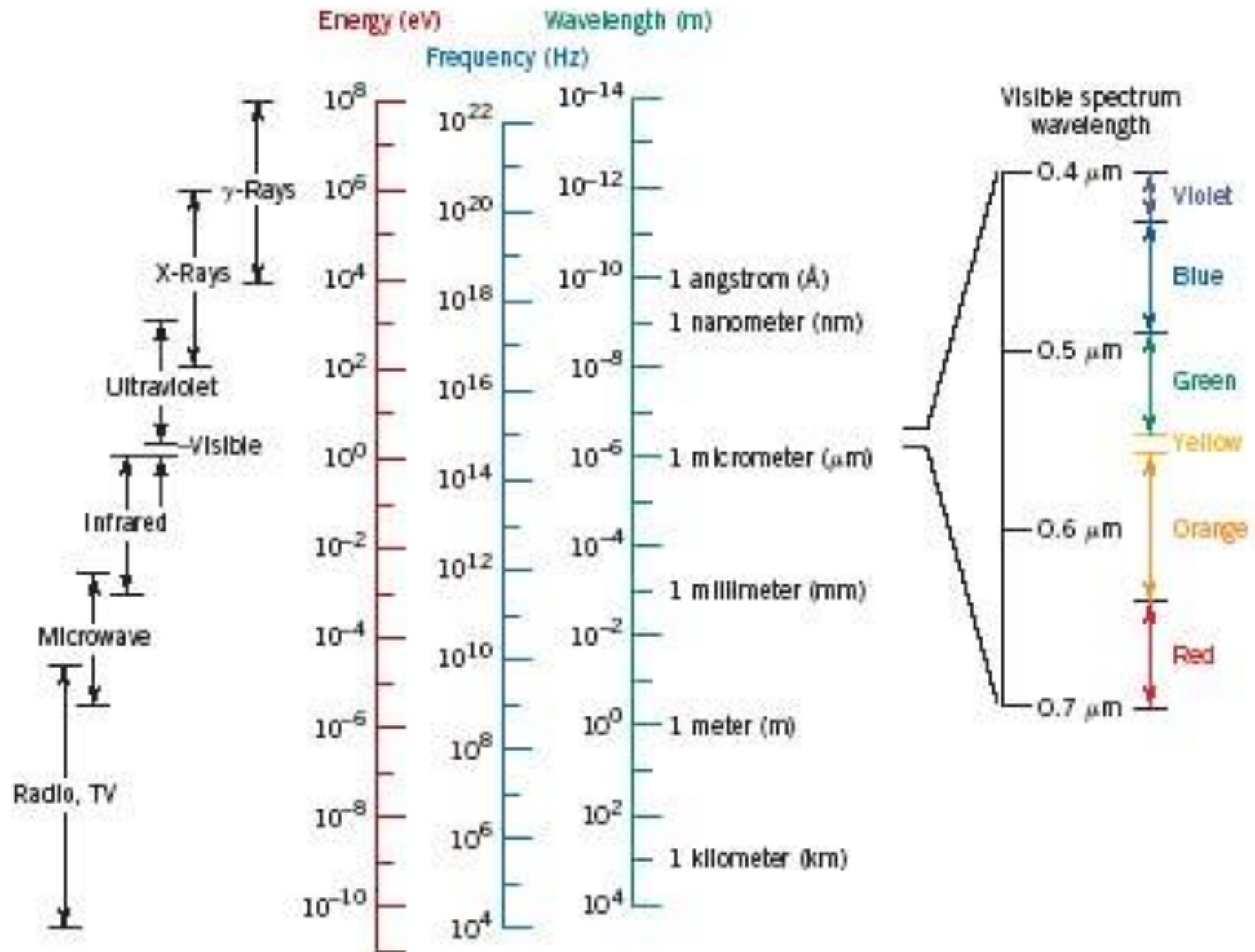
ν = frequency of radiation

h = Planck's constant ($6.62 \times 10^{-34} \text{ J} \cdot \text{s}$)

c = speed of light in a vacuum ($3.00 \times 10^8 \text{ m/s}$)



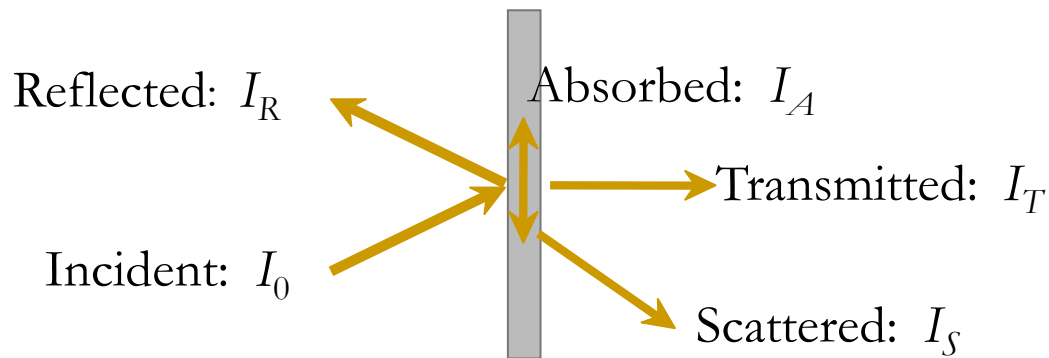
The spectrum of electromagnetic radiation



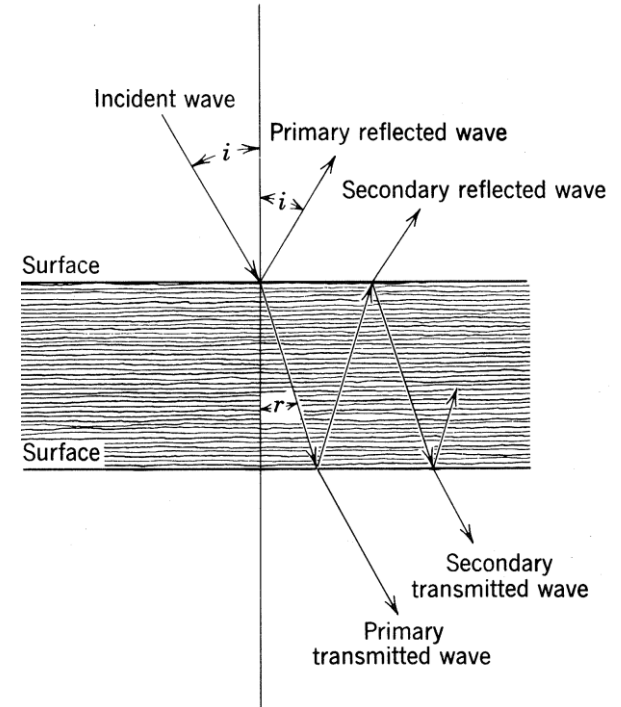
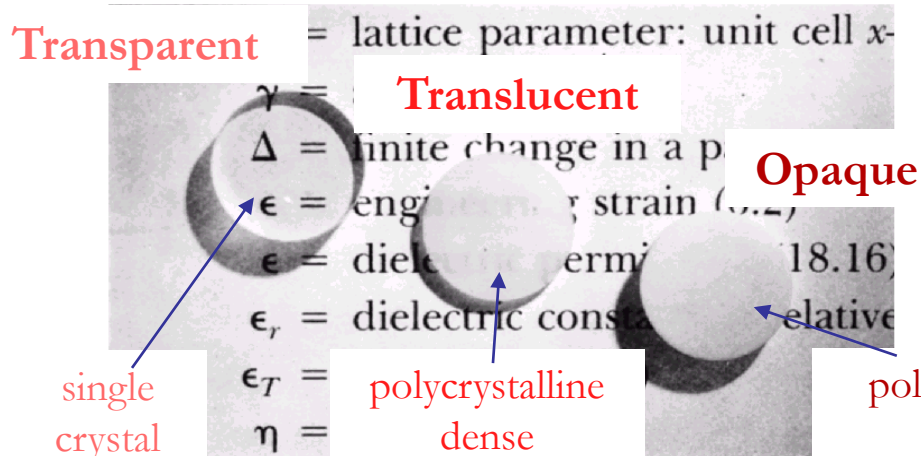
Light Interactions with Solids

- Incident light is reflected, absorbed, scattered, and/or transmitted:

$$I_0 = I_T + I_A + I_R + I_S$$



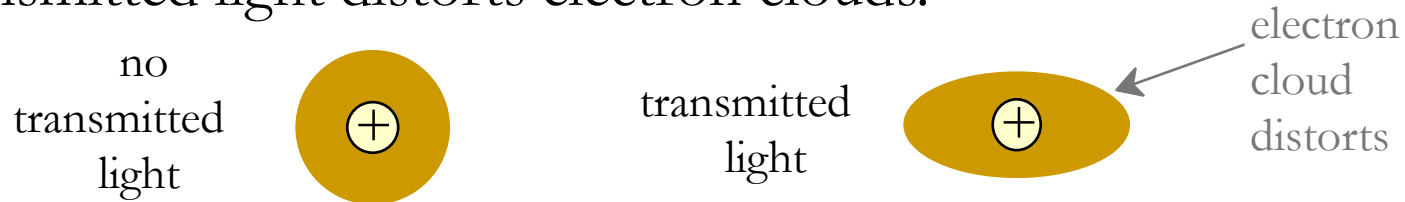
- Optical classification of materials:



Adapted from Fig. 21.10, *Callister 6e*.
(Fig. 21.10 is by J. Telford, with
specimen preparation by P.A. Lessing.)

Refractive Index

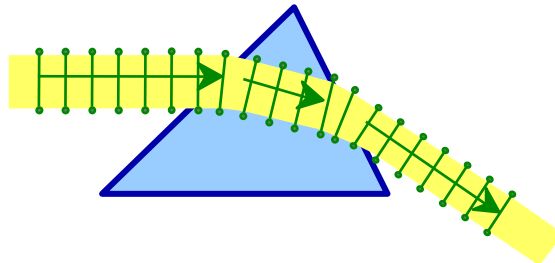
- Transmitted light distorts electron clouds.



- The velocity of light in a material is lower than in a vacuum.

$$n = \text{index of refraction} \equiv \frac{c \text{ (velocity of light in vacuum)}}{v \text{ (velocity of light in medium)}}$$

- Adding large ions (e.g., lead) to glass decreases the speed of light in the glass.
- Light can be “bent” as it passes through a transparent prism



Material	n
Typical glasses ca.	1.5 -1.7
Plastics	1.3 -1.6
PbO (Litharge)	2.67
Diamond	2.41

Selected values from Table 21.1,
Callister & Rethwisch 8e.

Factors influence Refractive Index

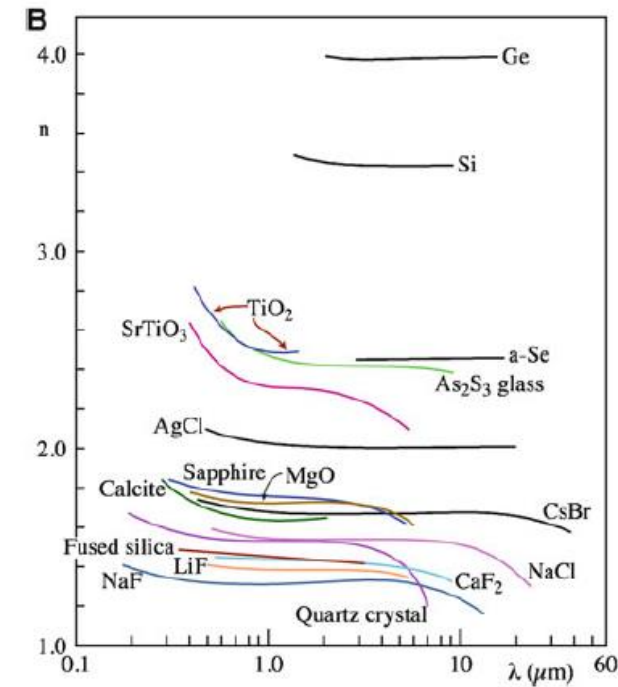
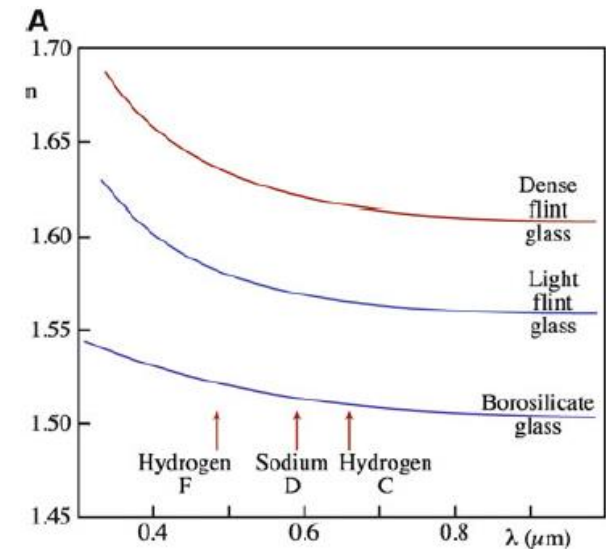
- **Ion size:** Materials containing large ions have large value of n ;

- **Structure:** Less-dense polymorphs have a more open structure and thus a lower n .

$$n_{\text{glass}} (1.46) < n_{\text{tridymite}} (1.47) < n_{\text{cristobalite}} (1.49) < n_{\text{quartz}} (1.55)$$

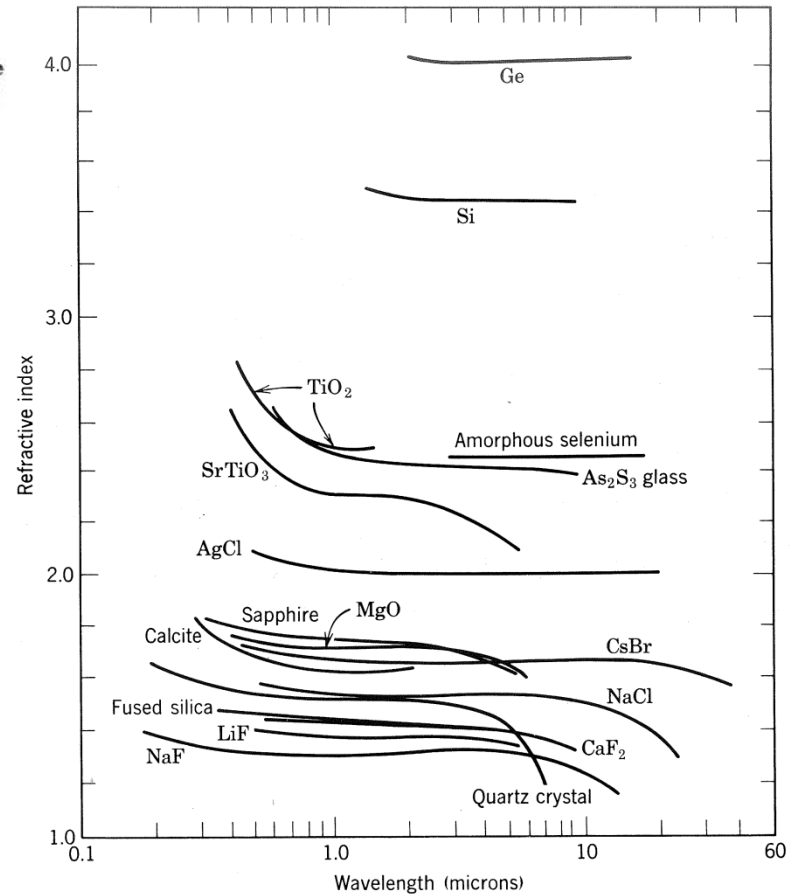
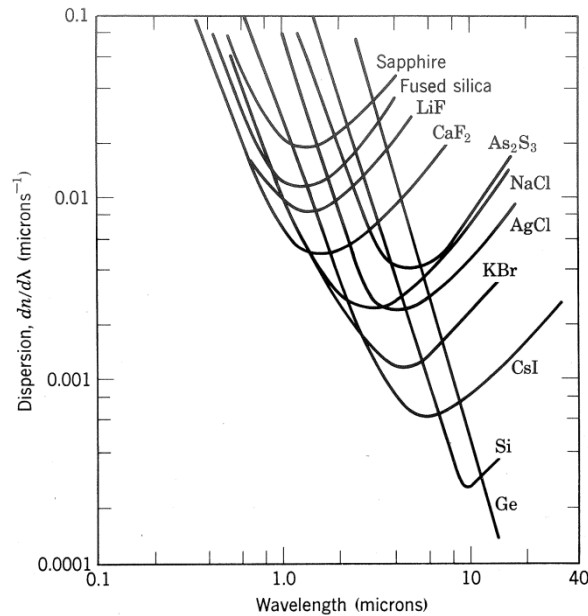
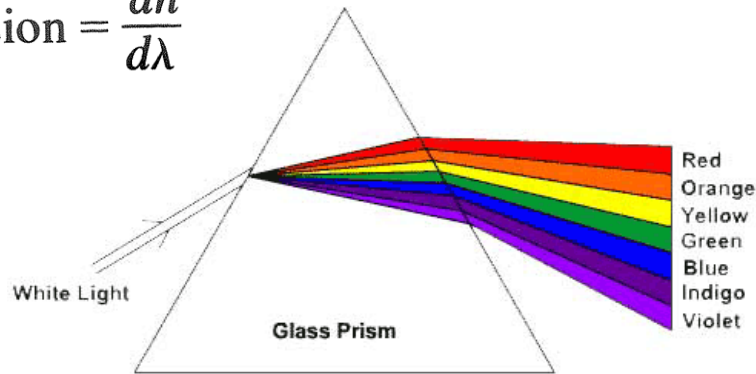
- **Crystallographic direction:** glasses and crystals with a cubic structure are optically isotropic. n is higher along close-packed directions.

Stress: tensile stress to an isotropic material causes an increase in n normal to the direction of the applied stress and decrease n along the stressed direction.



Dispersion

$$\text{Dispersion} = \frac{dn}{d\lambda}$$



- The refractive index is a function of the frequency of the light, normally decreasing as the wavelength increases.

Total Internal Reflectance

Snell's law:

$$\frac{n_1}{n_2} = \frac{\sin \phi_2}{\sin \phi_1} = \frac{v_2}{v_1}$$

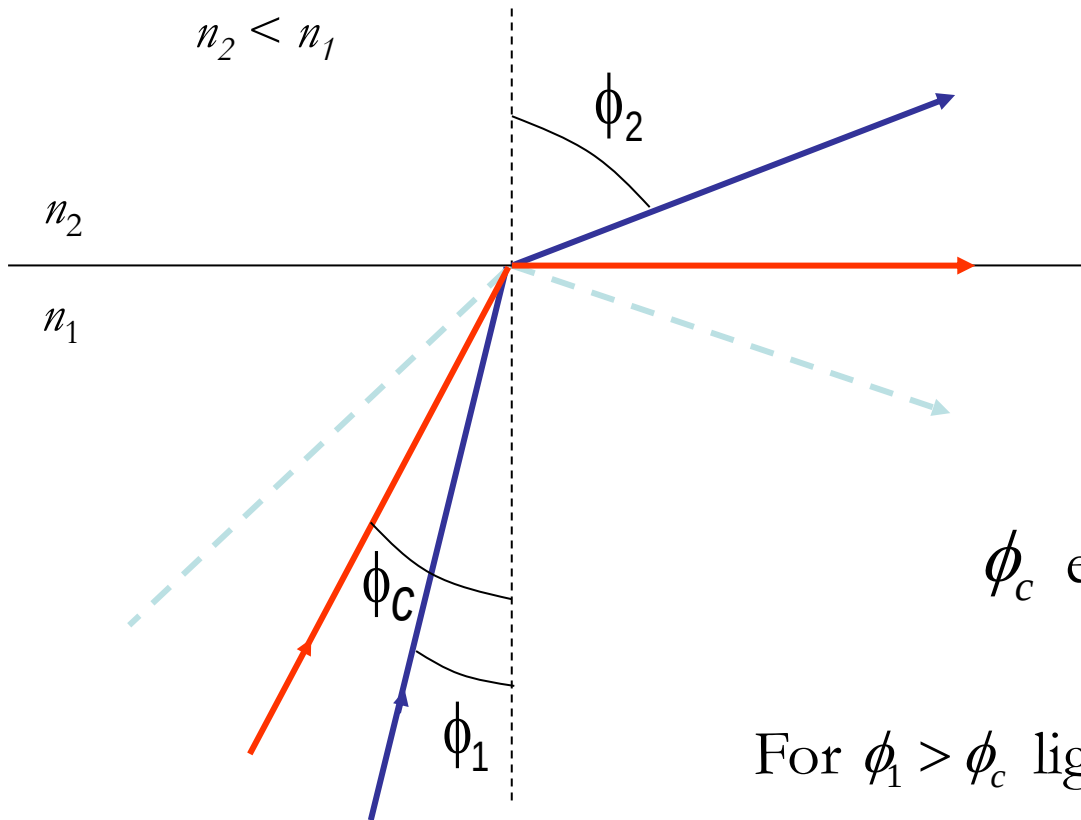
ϕ_1 = incident angle

ϕ_2 = refracted angle

ϕ_c = critical angle

ϕ_c exists when $\phi_2 = 90^\circ$

For $\phi_1 > \phi_c$ light is internally reflected



- Fiber optic cables are clad in low n material so that light will experience total internal reflectance and not escape from the optical fiber.

Reflectivity of Ceramics

- ✦ For normal incidence and light passing from vacuum/air into a solid having an index of refraction n :

$$R = \text{reflectivity} = \left(\frac{n-1}{n+1} \right)^2$$

- Example: For Diamond $n = 2.41$

$$R = \left(\frac{2.41-1}{2.41+1} \right)^2 = 0.17$$

$\therefore$ 17% of light is reflected

Example: Diamond in air

- What is the critical angle ϕ_c for light passing from diamond ($n_1 = 2.41$) into air ($n_2 = 1$)?
- Solution: At the critical angle,

and

$$\begin{aligned}\phi_1 &= \phi_c \\ \phi_2 &= 90^\circ\end{aligned}$$

Rearranging the equation

$$\frac{n_1}{n_2} = \frac{\sin \phi_2}{\sin \phi_1}$$

$$\sin \phi_1 = \sin \phi_c = \frac{n_2}{n_1} \sin(90^\circ) = \frac{n_2}{n_1}$$

Substitution gives

$$\sin \phi_c = \frac{1}{2.41}$$

$$\phi_c = 24.5^\circ$$

Light Absorption

The amount of light absorbed by a material is calculated using [Beer's Law](#)

$$\beta = 4\pi k / \lambda$$

$$I'_T = I'_0 e^{-\beta \ell}$$

β = absorption coefficient, cm^{-1}

ℓ = sample thickness, cm

I'_0 = incident light intensity

I'_T = transmitted light intensity

Rearranging and taking the natural log of both sides of the equation leads to

$$\ln \left[\frac{I'_T}{I'_0} \right] = -\beta \ell$$

Selected Light Absorption in Semiconductors

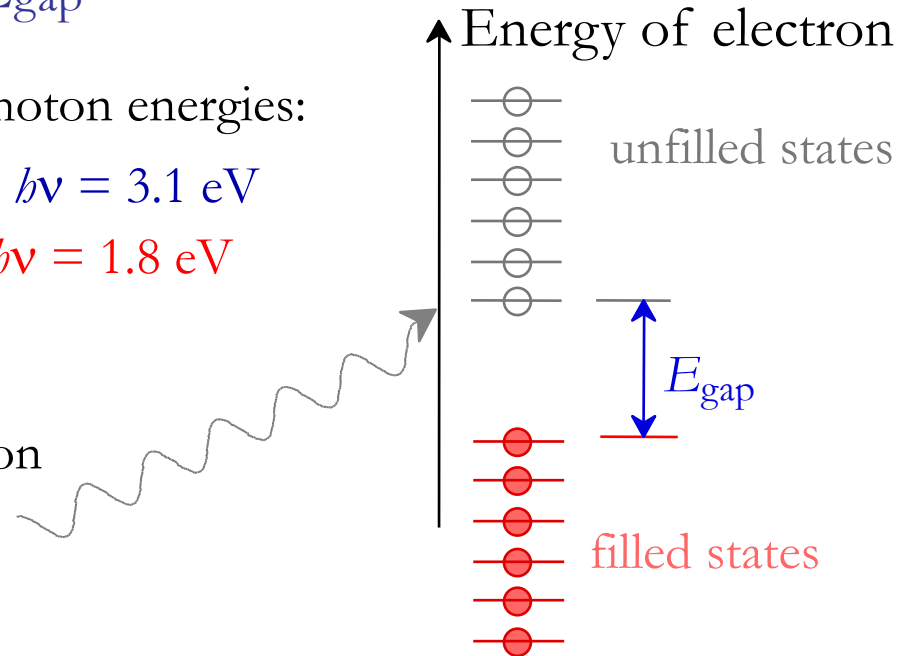
Absorption of light of frequency ν by electron transition occurs if $h\nu > E_{\text{gap}}$

Examples of photon energies:

violet light: $h\nu = 3.1 \text{ eV}$

red light: $h\nu = 1.8 \text{ eV}$

incident photon
energy $h\nu$



Adapted from Fig. 21.5(a), *Callister & Rethwisch 8e*.

- If $E_{\text{gap}} < 1.8 \text{ eV}$, all light absorbed; material is opaque (e.g., Si, GaAs)
- If $E_{\text{gap}} > 3.1 \text{ eV}$, no light absorption; material is transparent and colorless (e.g., diamond)
- If $1.8 \text{ eV} < E_{\text{gap}} < 3.1 \text{ eV}$, partial light absorption; material is colored

Computations of Minimum Wavelength Absorbed

- (a) What is the minimum wavelength absorbed by Ge, for which $E_g = 0.67$ eV?

Solution:

$$\lambda_{\text{Ge}}(\text{min}) = \frac{hc}{E_g(\text{Ge})} = \frac{(6.63 \times 10^{-34} \text{ J} \cdot \text{s})(3 \times 10^8 \text{ m/s})}{(0.67 \text{ eV})(1.60 \times 10^{-19} \text{ J/eV})}$$

$$\lambda_{\text{Ge}}(\text{min}) = 1.86 \times 10^{-6} \text{ m} = 1.86 \mu\text{m}$$

- (b) Redoing this computation for Si which has a band gap of 1.1 eV

$$\lambda_{\text{Si}}(\text{min}) = 1.13 \mu\text{m}$$

Note: the presence of donor and/or acceptor states allows for light absorption at other wavelengths.

Scattering

For optically heterogeneous systems, such as transparent medium containing small particles, part of the beam is scattered and the intensity is decreased.

For a single particle projected area πr^2 , the fraction of the beam intensity lost is given by

$$\frac{\Delta I}{I} = -K \frac{\pi r^2}{A}$$

Where r is the particle radius, A is the beam area, and K is a scattering factor that varies between 0 and 4. If we neglect multiple scattering,

$$\frac{I}{I_0} = \exp(-Sx) = \exp(-KN\pi r^2 x)$$

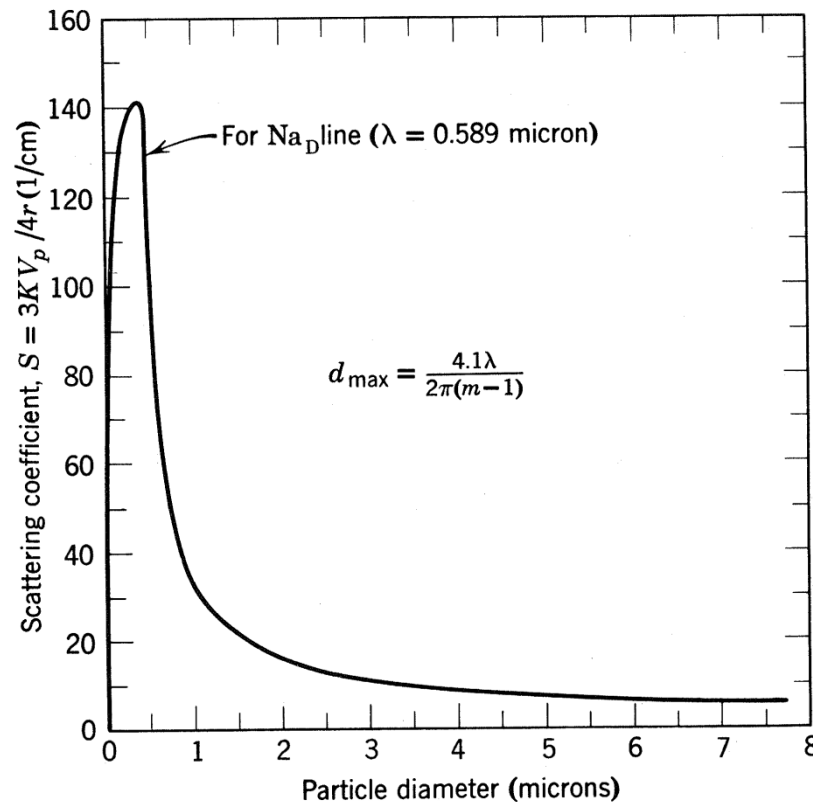
If the volume fraction of second phase particles present is V_p ,

$$V_p = \frac{4}{3} \pi r^3 \cdot N$$

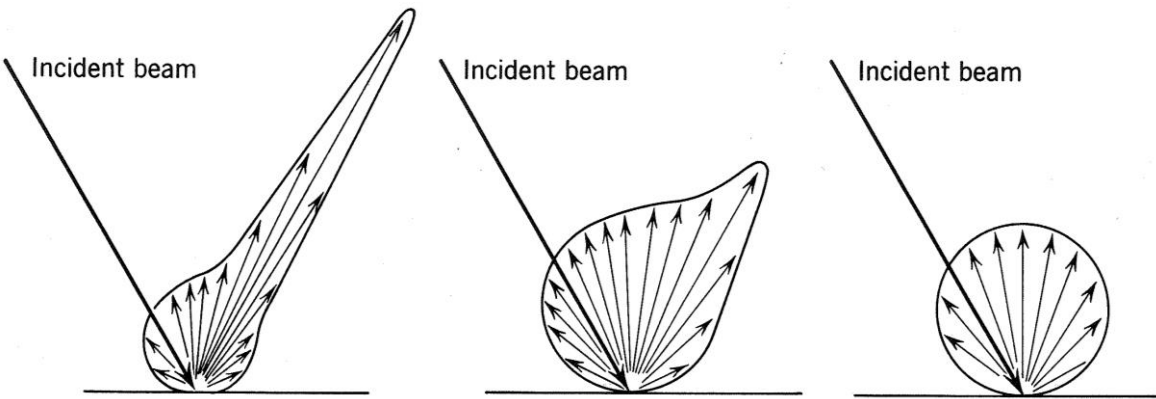
$$\frac{I}{I_0} = \exp\left(-3KV_p \frac{x}{4r}\right)$$

Scattering coefficient K

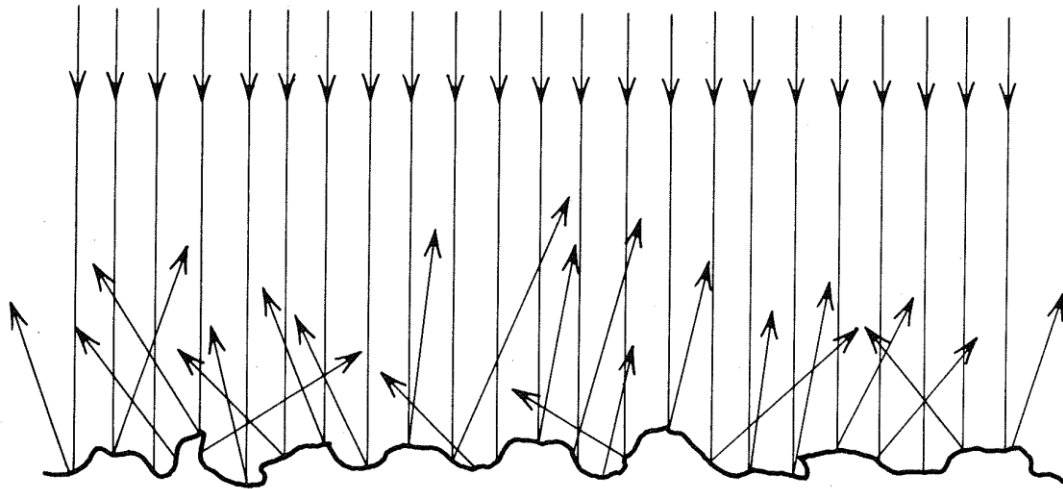
The scattering factor K is mainly dependent on (1) the ratio of the particle diameter to the wavelength of the incident light ($\alpha = 2\pi r / \lambda$); (2) the relative refractive index of the particle and the medium ($m = n_{\text{particle}} / n_{\text{medium}}$)



Surface gloss



Polar diagrams of reflection from surfaces of increasing roughness



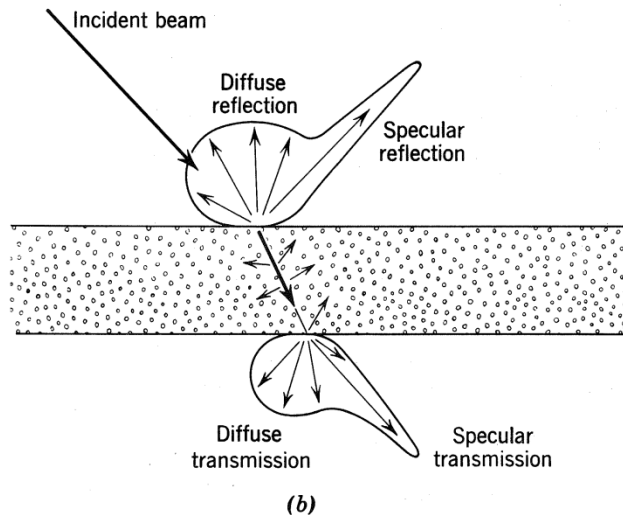
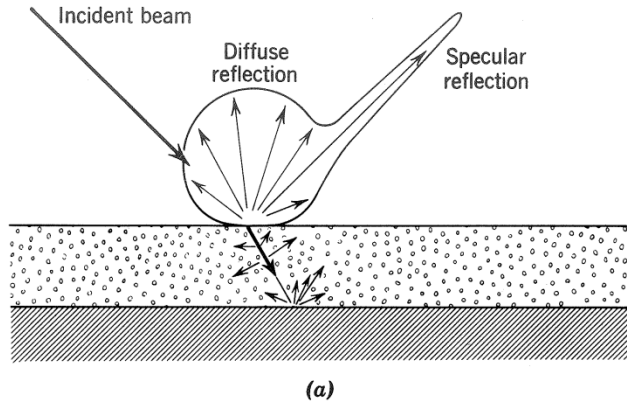
Reflection from a rough surface

The gloss of a surface is related to the relative amount of specular and diffuse reflections.

Refractive index and surface topography

Difficulties in obtaining high gloss with glazes and enamels usually result from surface roughness caused by crystal formation, a wavy surface, or craters caused by bursting bubbles.

Opacity



Specular and diffuse reflection and transmission from (a) surface coating of glaze or enamel and (b) plate of translucent glass or procelain.

The main factors determining the overall scattering coefficient and consequently affecting the opacity of a two-phase system are the particle size, relative refractive index, and volume of second phase particles present.

Opacifiers	n_D	$n_{\text{crystal}}/n_{\text{glass}}$
Inert additions		
SnO_2	2.0	1.33
ZrSiO_4	2.0	1.33
ZrO_2	2.2	1.47
ZnS	2.4	1.6
TiO_2 (anatase)	2.52	1.68
TiO_2 (rutile)	2.76	1.84
Inert products of reactions on melting		
Air Pores	1.0	0.67
As_2O_5	2.2	1.47
PbAs_2O_6	2.2	1.47
$\text{Ca}_4\text{Sb}_4\text{O}_{13}\text{F}_2$	2.2	1.47
Nucleated and crystallized from the glass		
NaF	1.3	0.87
CaF_2	1.4	0.93
CaTiSiO_5	1.9	1.27
ZrO_2	2.2	1.47
CaTiO_3	2.35	1.57
TiO_2 (anatase)	2.52	1.68
TiO_2 (rutile)	2.76	1.84

Translucency

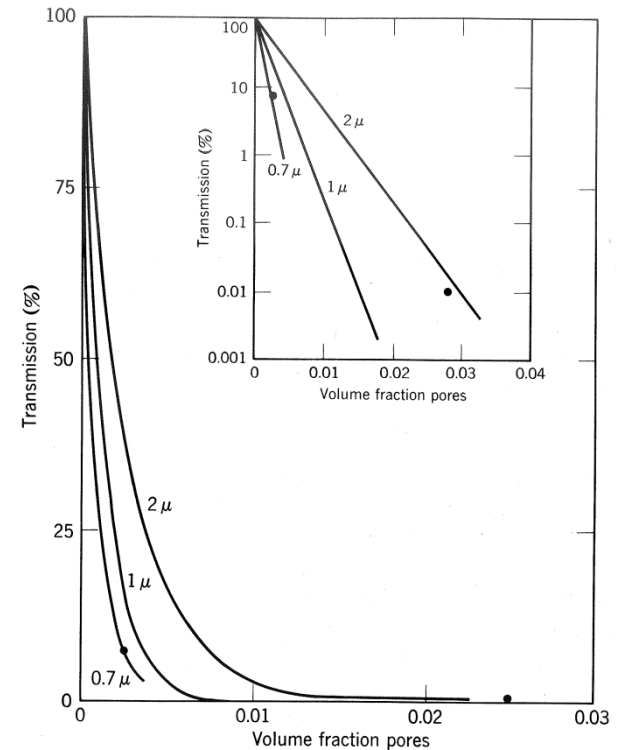
Translucency is a super-set of transparency: it allows light to pass through, but does not necessarily follow Snell's law; the photons can be scattered at either of the two interfaces where there is a change in index of refraction, or internally.

Opal glass/translucent porcelain:

Appreciable scattering with a minimum absorption;

Maximum of diffuse transmission;

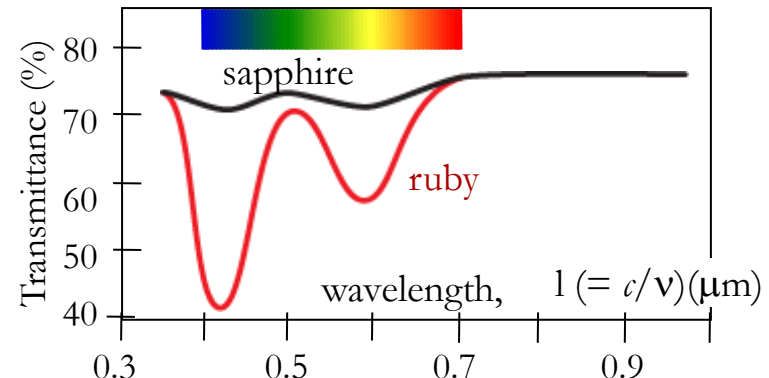
A dispersed phase with an index not much different from that of the matrix is most satisfactory.



Transmission of polycrystalline alumina containing small amounts of residual porosity

Color of Ceramics

- Color determined by the distribution of wavelengths:
 - transmitted light
 - re-emitted light from electron transitions
- Example 1: Cadmium Sulfide (CdS), $E_g = 2.4 \text{ eV}$
 - absorbs higher energy visible light (blue, violet)
 - color results from red/orange/yellow light that is transmitted
- Example 2: Ruby = Sapphire (Al_2O_3) + (0.5 to 2) at% Cr_2O_3
 - Sapphire is transparent and colorless ($E_g > 3.1 \text{ eV}$)
 - adding Cr_2O_3 :
 - alters the band gap
 - blue and orange/yellow/green light is absorbed
 - red light is transmitted
 - Result: Ruby is deep red in color

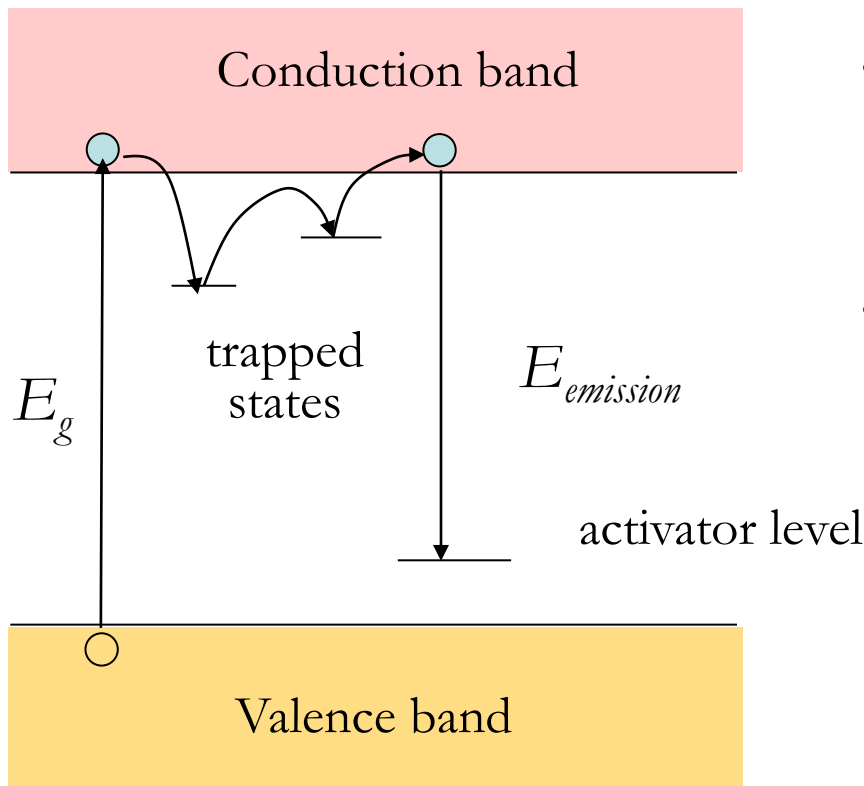


Adapted from Fig. 21.9, *Callister & Rethwisch 8e*. (Fig. 21.9 adapted from "The Optical Properties of Materials" by A. Javan, *Scientific American*, 1967.)

Luminescence

📌 Luminescence – reemission of light by a material

- 📄 Material absorbs light at one frequency and reemits it at another (lower) frequency.
- 📄 Trapped (donor/acceptor) states introduced by impurities/defects

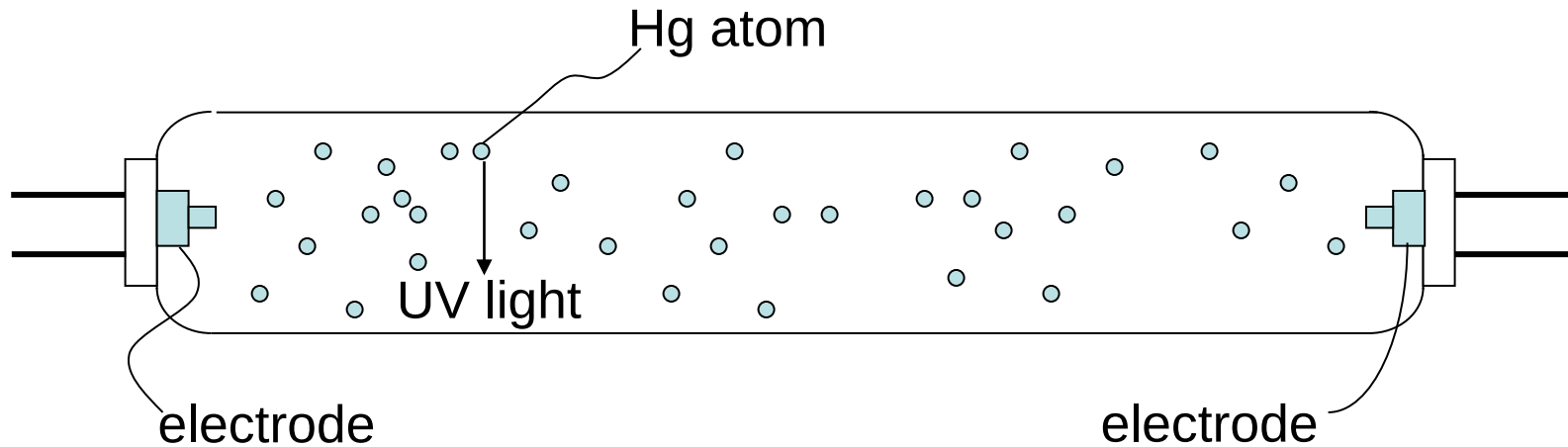


- If residence time in trapped state is relatively long ($> 10^{-8}$ s)
-- phosphorescence

- For short residence times ($< 10^{-8}$ s)
-- fluorescence

Example: Toys that glow in the dark.
Charge toys by exposing them to light.
Reemission of light over time—
phosphorescence

Photoluminescence



- ✦ Arc between electrodes excites electrons in mercury atoms in the lamp to higher energy levels.
- ✦ As electron falls back into their ground states, UV light is emitted (e.g., suntan lamp).
- ✦ Inside surface of tube lined with material that absorbs UV and reemits visible light
 - For example, $\text{Ca}_{10}\text{F}_2\text{P}_6\text{O}_{24}$ with 20% of F^- replaced by Cl^-
- ✦ Adjust color by doping with metal cations

Sb^{3+} blue

Mn^{2+} orange-red

Cathodoluminescence (阴极发光)

- ✦ Used in cathode-ray tube devices (e.g., TVs, computer monitors)
 - ✦ Inside of tube is coated with a phosphor material
 - 📄 Phosphor material bombarded with electrons
 - 📄 Electrons in phosphor atoms excited to higher state
 - 📄 Photon (visible light) emitted as electrons drop back into ground states
 - 📄 Color of emitted light (i.e., photon wavelength) depends on composition of phosphor
- | | |
|--|-------|
| $\text{ZnS (Ag}^+ \text{ \& Cl}^-)$ | blue |
| $(\text{Zn, Cd}) \text{ S} + (\text{Cu}^+ + \text{Al}^{3+})$ | green |
| $\text{Y}_2\text{O}_2\text{S} + 3\% \text{ Eu}$ | red |
- ✦ Note: light emitted is random in phase & direction
 - 📄 i.e., is noncoherent

The LASER

- ✦ The laser generates light waves that are in phase (coherent) and that travel parallel to one another

LASER

 Light Amplification by Stimulated Emission of Radiation

- ✦ Operation of laser involves a **population inversion** of energy states process

Population Inversion

- More electrons in excited energy states than in ground states

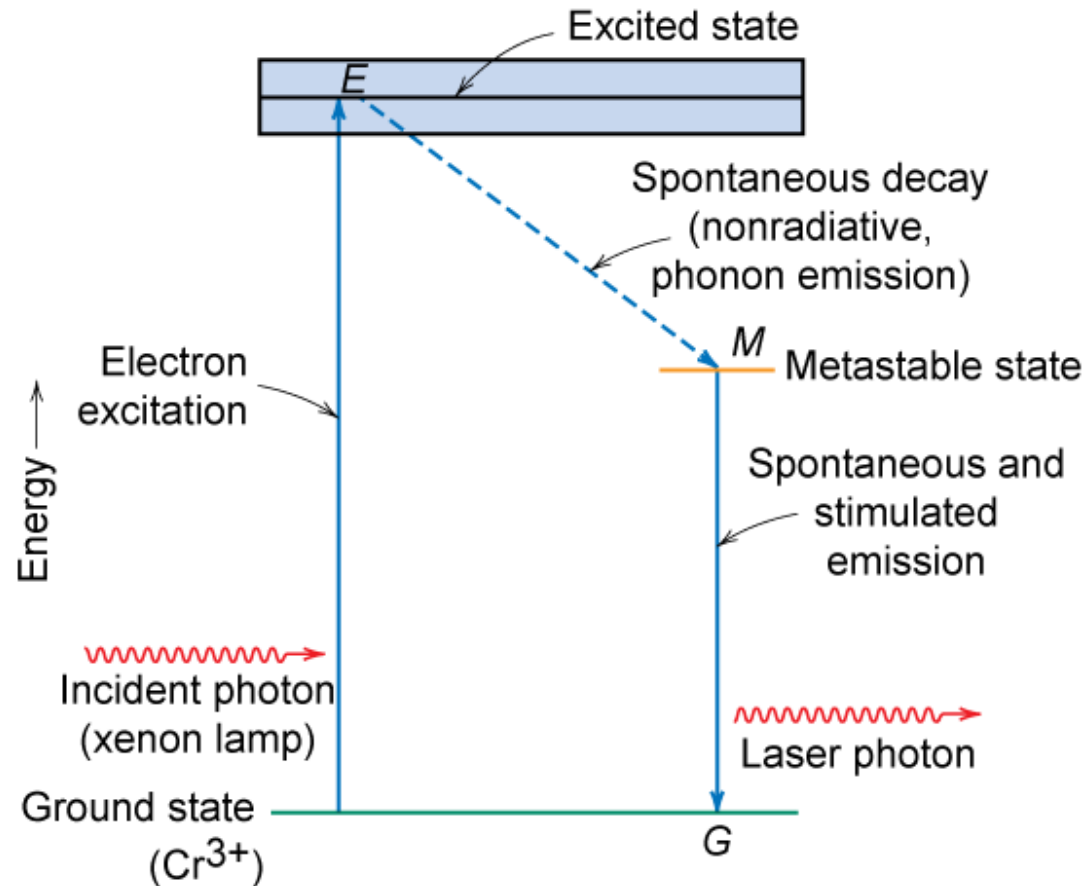


Fig. 21.14, *Callister & Rethwisch 8e.*

Operation of the Ruby Laser

- ✦ “pump” electrons in the lasing material to excited states
 - 📄 e.g., by flash lamp (incoherent light).

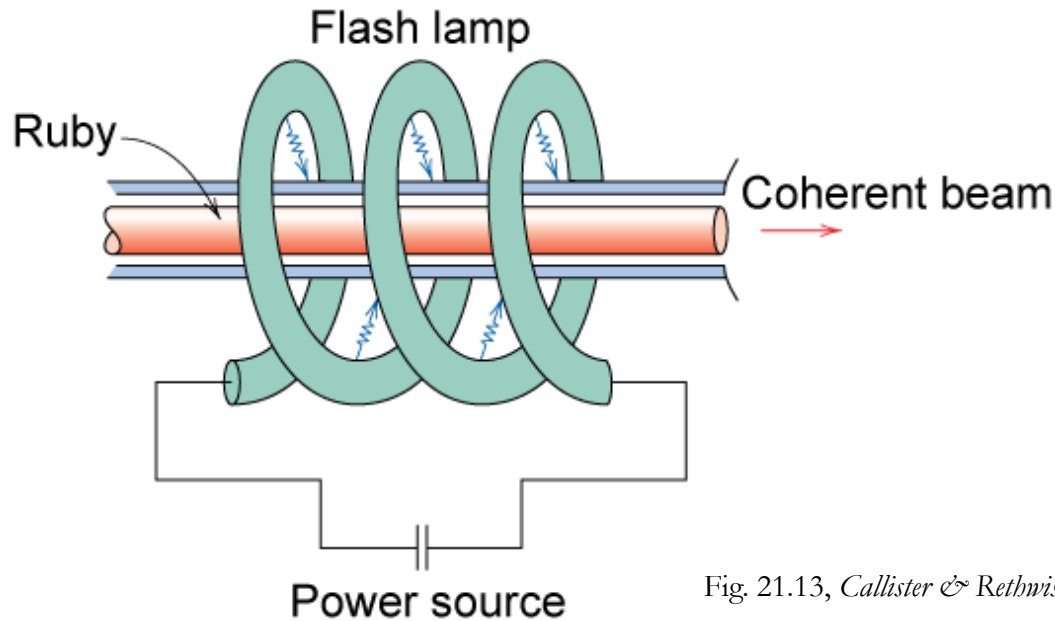


Fig. 21.13, *Callister & Rethwisch 8e.*

- Direct electron decay transitions — produce incoherent light

Operation of the Ruby Laser (cont.)

Stimulated Emission

- ☞ The generation of one photon by the decay transition of an electron, induces the emission of other photons that are all in phase with one another.
- ☞ This cascading effect produces an intense burst of coherent light.

☞ This is an example of a **pulsed laser**

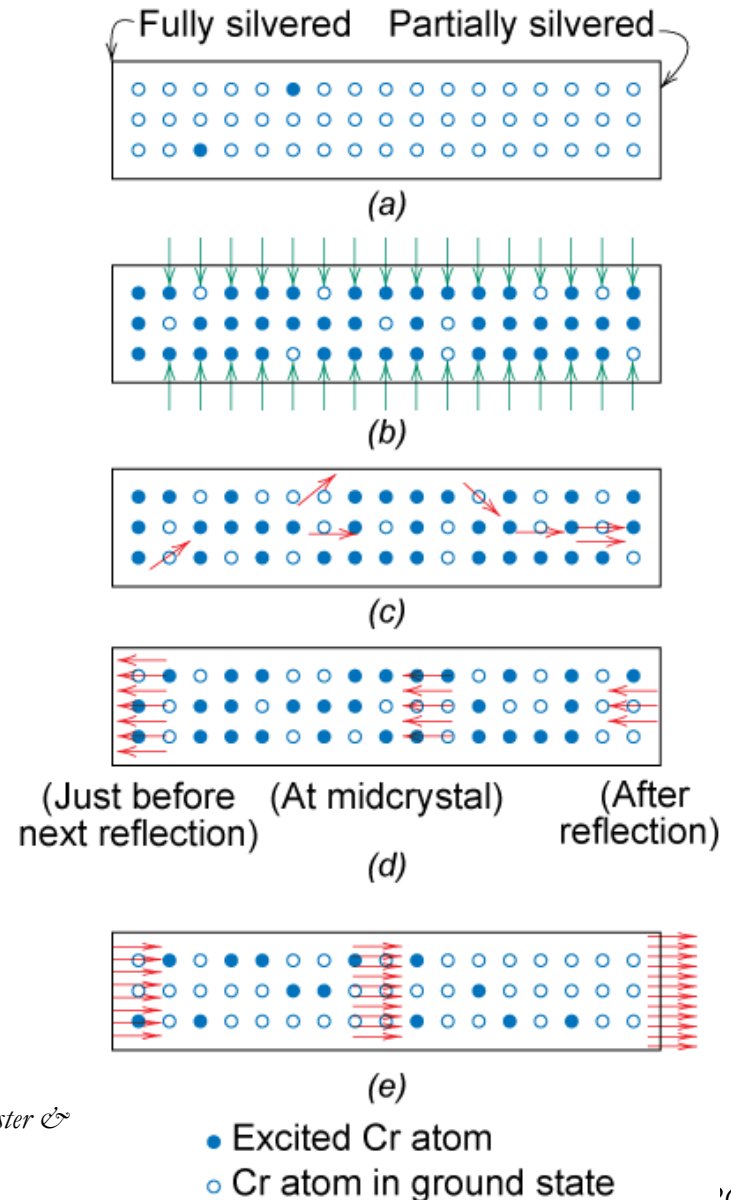


Fig. 21.15, *Callister & Rethwisch 8e.*

Continuous Wave Lasers

- ✚ Continuous wave (CW) lasers generate a continuous (rather than pulsed) beam
- ✚ Materials for CW lasers include semiconductors (e.g., GaAs), gases (e.g., CO₂), and yttrium-aluminum-garnet (YAG)
- ✚ Wavelengths for laser beams are within visible and infrared regions of the spectrum
- ✚ Uses of CW lasers
 1. Welding
 2. Drilling
 3. Cutting – laser carved wood, eye surgery
 4. Surface treatment
 5. Scribing – ceramics, etc.
 6. Photolithography – Excimer laser

Semiconductor Laser Applications

- ❖ Apply strong forward bias across semiconductor layers, metal, and heat sink.
- ❖ Electron-hole pairs generated by electrons that are excited across band gap.
- ❖ Recombination of an electron-hole pair generates a photon of laser light

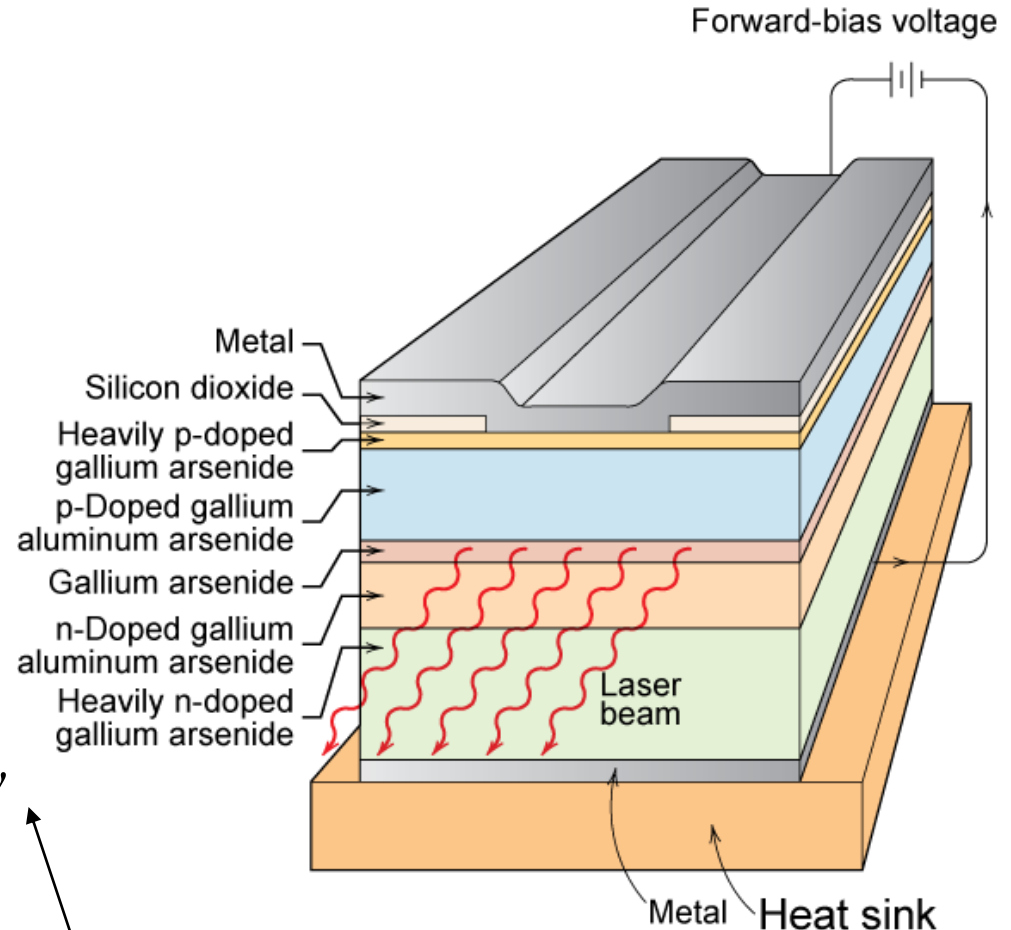
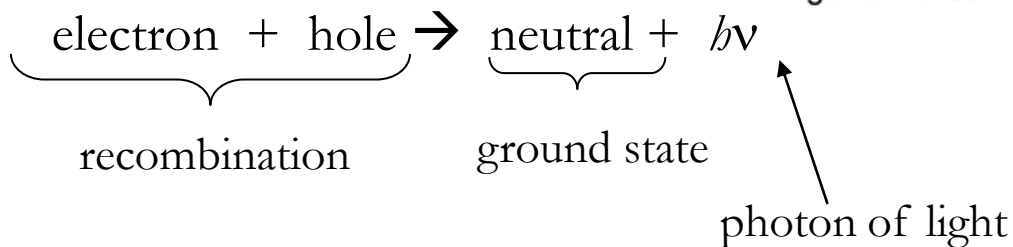


Fig. 21.17, Callister & Rethwisch 8e.

Semiconductor Laser Applications

- ✦ Compact disk (CD) player

- 📄 Use red light

- ✦ High resolution DVD players

- 📄 Use blue light

- 📄 Blue light is a shorter wavelength than red light so it produces higher storage density

- ✦ Communications using optical fibers

- 📄 Fibers often tuned to a specific frequency

- ✦ Banks of semiconductor lasers are used as flash lamps to pump other lasers

Other Applications of Optical Phenomena

- ✦ New materials must be developed to make new & improved optical devices.

- 📄 Organic Light Emitting Diodes (OLEDs)

- 📖 More than one color available from a single diode

- 📖 Also sources of white light (multicolor)

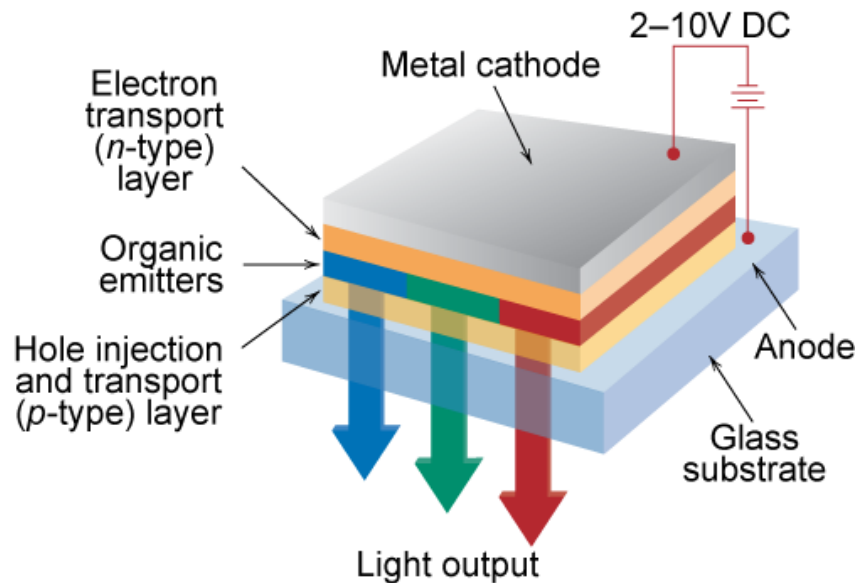
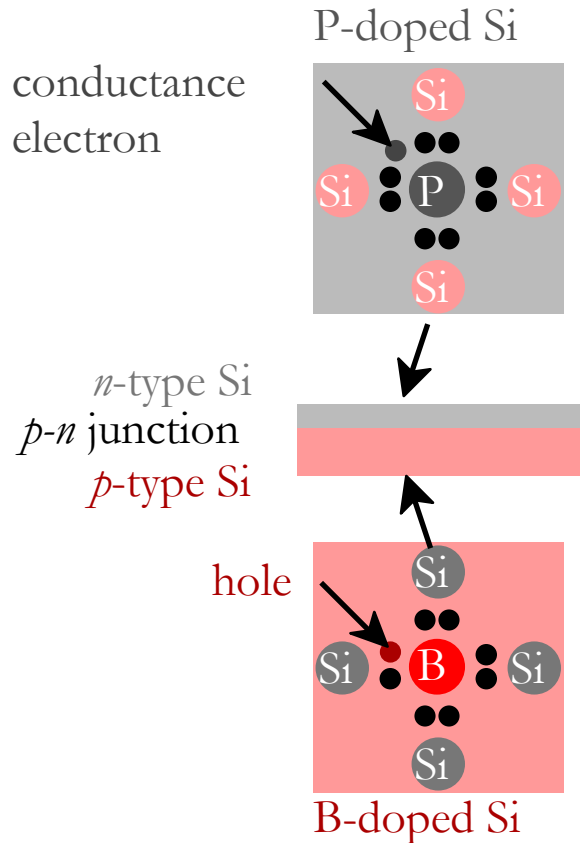


Fig. 21.12, *Callister & Rethwisch 8e*.
(Reproduced by arrangement with *Silicon Chip* magazine.)

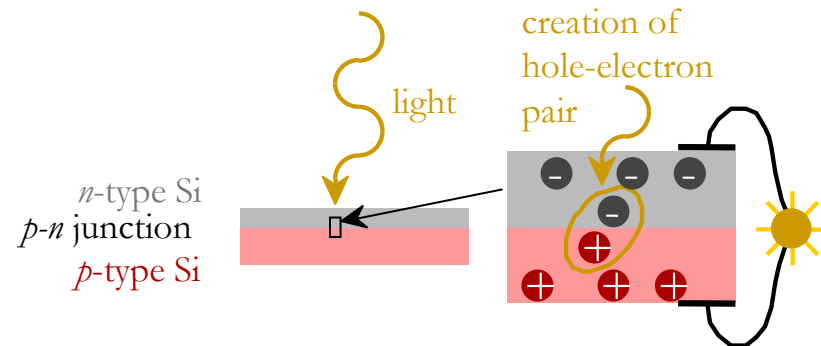
Other Applications - Solar Cells

- p - n junction:

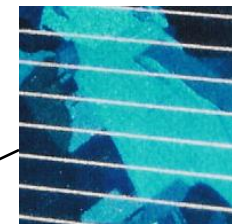


- Operation:

- incident photon of light produces elec.-hole pair.
- typical potential of 0.5 V produced across junction
- current increases w/light intensity.



- Solar powered weather station:



polycrystalline Si

Los Alamos High School weather station (photo courtesy P.M. Anderson)

Other Applications - Optical Fibers

Schematic diagram showing components of a fiber optic communications system

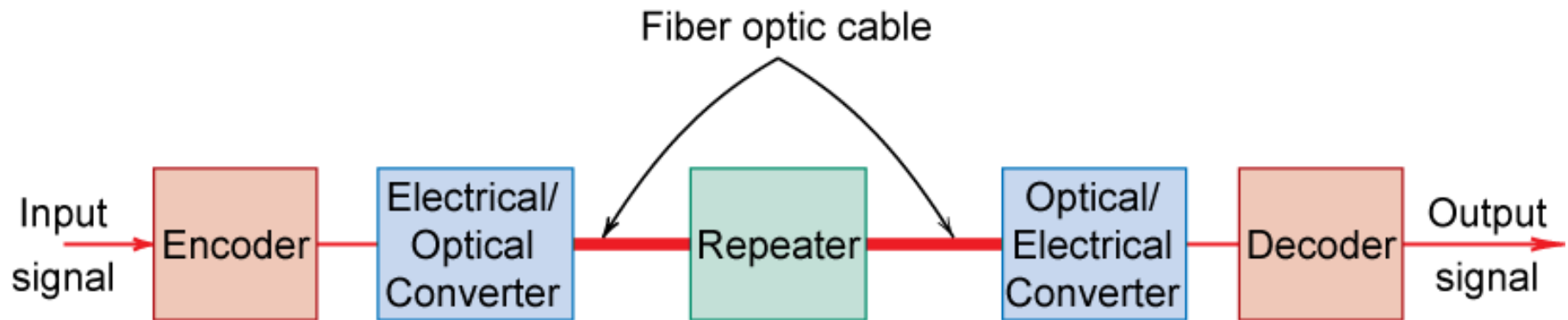


Fig. 21.18, Callister & Rethwisch 8e.

Optical Fibers (cont.)

- ✦ fibers have diameters of 125 μm or less
- ✦ plastic cladding 60 μm thick is applied to fibers

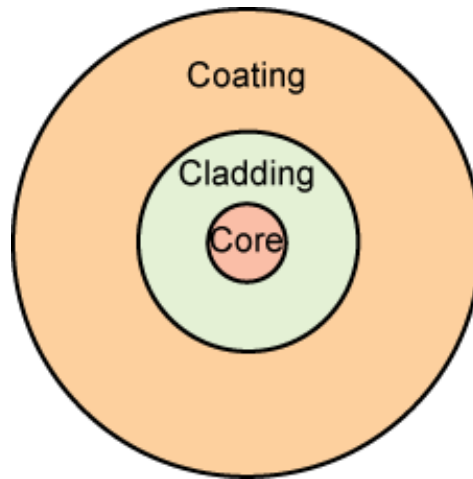


Fig. 21.20, *Callister & Rethwisch 8e.*

Optical Fiber Designs

Step-index Optical Fiber

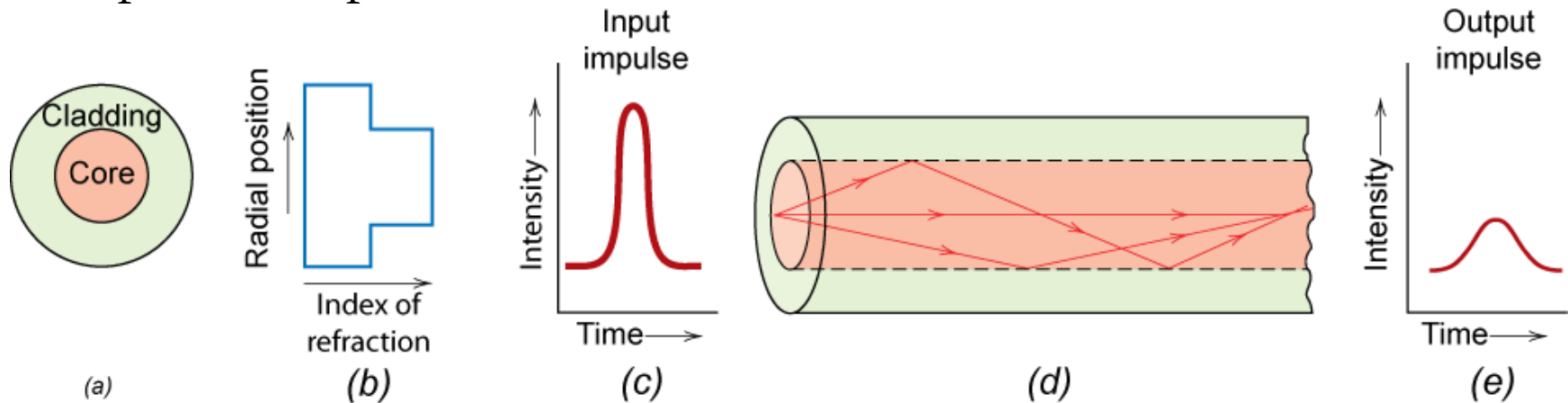


Fig. 21.21, Callister & Rethwisch 8e.

Graded-index Optical Fiber

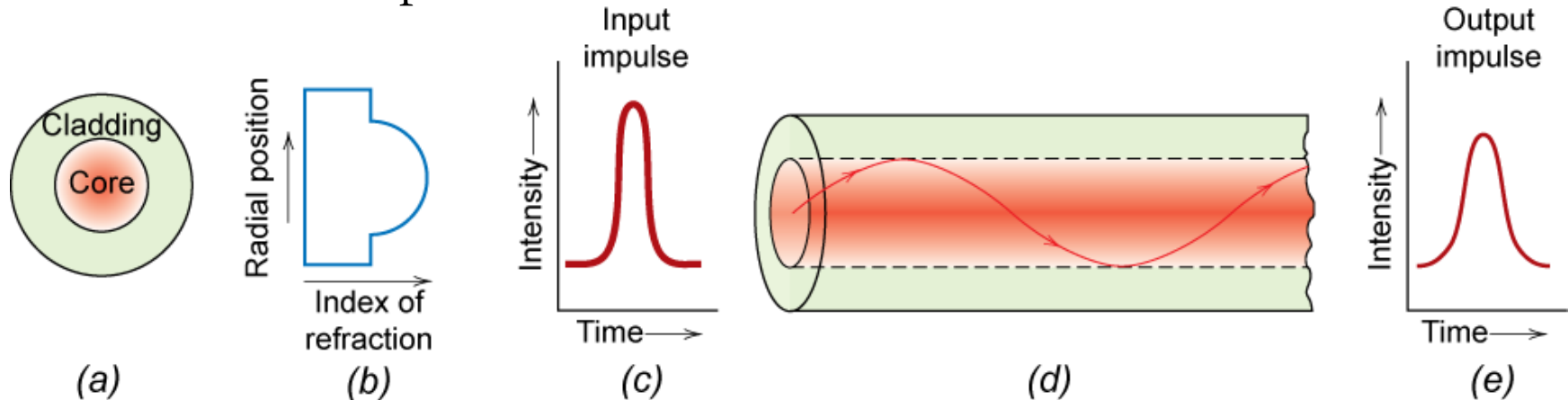


Fig. 21.22, Callister & Rethwisch 8e.

SUMMARY

- Light radiation impinging on a material may be reflected from, absorbed within, and/or transmitted through
- Light transmission characteristics:
 - transparent, translucent, opaque
- Optical properties of Ceramics:
 - for $E_{gap} < 1.8$ eV, absorption of all wavelengths of light radiation
 - for $E_{gap} > 3.1$ eV, no absorption of visible light radiation
 - for $1.8 \text{ eV} < E_{gap} < 3.1$ eV, absorption of some range of light radiation wavelengths
 - color determined by wavelength distribution of transmitted light
- Other important optical applications/devices:
 - luminescence, photoconductivity, light-emitting diodes, solar cells, lasers, and optical fibers